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# Image Analysis Protocol for Pollen Viability Identification on Selected Genotypes of Rice (*Oryza sativa* L.)

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***Abstract*—Pollen viability plays a key role in the efficient sexual reproduction of plants. However, estimating pollen viability is a labor-intensive and time-consuming task for researchers. Moreover, it is difficult to evaluate a large dataset of microscopic images as most of it needs to be meticulously examined. Existing methods have been developed to efficiently estimate total pollen count but their success in computing viability percentage is rather limited since they do not integrate viability testing techniques such as pollen staining. Some of these existing methods greatly depend on features such as the diameter of the pollen grain which requires a manual standardization to come up with two exclusive distributions for viable and nonviable pollen grains. An image analysis protocol was presented to compute the viability percentage on selected genotypes of rice (*Oryza sativa* L.). The effectiveness of the segmentation was evaluated in terms of accuracy in classification of the pollen grains. Results suggest the proposed method can be an alternative to the traditional labor-intensive manual counting.**

***Keywords-edge detection; k-means clustering; pollen viability; oryza sativa L.***

## I. INTRODUCTION

Rice is the most important crop for human consumption which provides staple food for more than half of the world's population [1]. Various researches on rice such as rice salt tolerance involves assessment of viability of its pollen. Pollen viability analysis is essential to provide further insights on relative responses of rice genotypes to biotic and abiotic stress [2]. It is imperative for a range of plant biologists from plant breeders to evolutionary biologists to be familiar with the extent of the viability of a pollen sample to be used in their plant pollination or experimentation in order to have an effective research. Determining the viability of a pollen also helps in efficient monitoring of crops in the case of breeders and growers. However, traditional techniques such as manual counting and assessment is labor-intensive and consumes a lot of time. Automated pollen viability identification is beneficial providing alternative techniques to make this task easier.

A pollen is a powdery substance produced by the anthers of seed-bearing plants, consisting of numerous fine pollen grains containing male gametes - i.e. germ cells that fuse with other cells during fertilization, which occurs during the plant pollination process [3]. These pollen grains are microscopic structures as shown in Figure 1, which is mostly responsible in the fertilization of a plant as it represents the male portion of the reproductive process in plants.

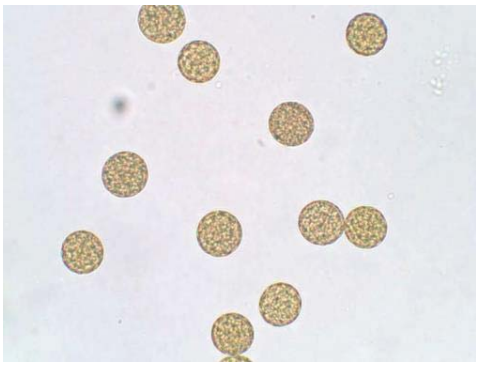


Figure 1. A microscopic image of a pollen.

In this study, a combination of image processing methods was utilized to automate pollen viability classification. The proposed approach includes Edge detection, Dilation, and Region Filling. These series of image processing techniques applied to input digital image determines the area in the image that are classified as pollens. *K*-means algorithm, which is an unsupervised learning and clustering algorithm, was employed to classify viable and non-viable pollens in the image.

## II. PRELIMINARIES

In this section, important algorithms will be covered to help in better understanding the automated viability classification process from which several concepts the proposed approach are based.

### *A. Edge Detection Algorithm*

Image processing is a method that performs operations on an image to enhance the image or extract useful details from it for further analysis and/or identification. Image processing plays a vital role in this study as there is a need to extract the features of the pollen grains from the image. The edge-based segmentation methods attempt to resolve image segmentation by detecting the edges or pixels between different regions that have a rapid change of intensity value linked to form closed object boundaries [4]. Some of the basic edge detection available are Canny operator, Sobel operator, and Prewitt operator.

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*1) Canny operator*

Canny edge detection is a multistage algorithm [5] [6]. The algorithm used in Canny edge detection has four major steps as follows:

- Apply Gaussian filter to the image
- Calculate the value and direction of the gradients
- Non-maxima suppression to gradient value
- Hysteresis thresholding to check real and nonreal edges

*2) Sobel operator*

The Sobel edge detector computes the gradient by using the discrete differences between rows and columns of a 3×3 neighborhood [7]. The operator consists of a pair of 3×3 convolution kernels, as shown in Figure 2 which are used to detect edges in the vertical and horizontal direction respectively.

$$G_x = \begin{bmatrix} -1 & 0 & +1 \\ -2 & 0 & +2 \\ -1 & 0 & +1 \end{bmatrix} \qquad G_y = \begin{bmatrix} +1 & +2 & +1 \\ 0 & 0 & 0 \\ +1 & +2 & +1 \end{bmatrix}$$

Figure 2. 3×3 Sobel operator.

*3) Prewitt operator*

Prewitt operator edge detection masks are one of the oldest and best-understood methods of detecting edges in images. The Prewitt edge detector uses the following mask to approximate digitally the first derivatives $G_x$ and $G_y$ [7] as shown in Figure 3.

$$G_x = \begin{bmatrix} -1 & 0 & +1 \\ -1 & 0 & +1 \\ -1 & 0 & +1 \end{bmatrix} \qquad G_y = \begin{bmatrix} +1 & +1 & +1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix}$$

Figure 3. 3×3 Sobel operator.

## B. k-Means Clustering

Clustering is the process of subdividing a group of data into smaller subsets. Generally, we have n data points $x_i$, $i = 1 \ldots n$ that have to be subdivided into $k$ groups. *K*-means clustering method aims at finding positions $u_i$, $i = 1 \ldots k$ of the clusters that minimize the distance from the data points to the cluster. *K*-means typically solves an objective function known as the squared error function given by:

$$\arg\min_{c} \sum_{i=1}^{k} \sum_{x \in c_i} d(x, \mu_i) = \arg\min_{c} \sum_{i=1}^{k} \sum_{x \in c_i} \|x - \mu_i\|_2^2 \qquad (1)$$

where $c_i$ is the set of points that belong to cluster $i$. The *k*-means clustering uses the square of the Euclidean distance $d(x,u_i)=||x-u_i||^2$ [8].

The *k*-means clustering algorithm uses an iterative scheme to produce a minimal solution. While there are a number of variants to this algorithm, Lloyd's algorithm will be used in this study. The algorithm requires as input a dataset and the number of clusters $k$ then executes the following steps:

1. Initialize estimates for k-centroids. This can either be randomly generated or randomly selected from the dataset.
2. Assign each data point to the closest centroid.
3. Update centroid. In this step, the centroid is recalculated. This is done by taking the mean of all data points assigned to that centroid's cluster
4. Repeat steps 2 and 3 until convergence is achieved.

The algorithm will eventually converge to a point. The result may not be the best possible outcome. That is because the problem is non-convex and the algorithm is just heuristic, converging to a local optimum. The algorithm will terminate if the assignments remain unchanged from previous to the current iteration [8].

# III. Proposed Image Analysis Protocol for Pollen Viability Identification

The overall flow of the proposed approach for pollen viability classification is shown in Figure 4.

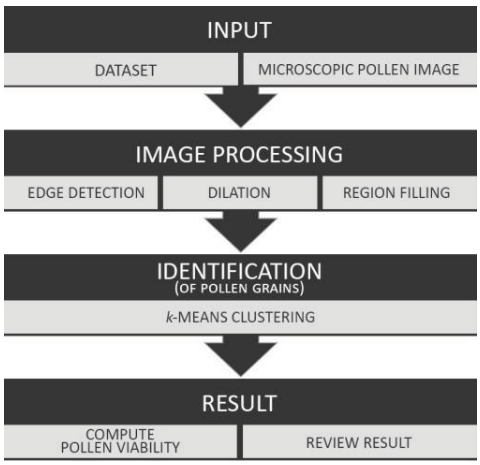



Figure 4. Workflow of proposed approach.

## A. Input

The dataset used in this study is composed of microscopic pollen images of selected genotypes of rice (*Oryza sativa* L.) with staining method applied to it. The rice genotypes collected and used were IR68652-3B-8-1, Sadri, CSR28, IR67075-2B-5-2, Rc222 and Pokkali.

## B. Image Processing

A combination of image processing techniques was applied to the dataset images to be able to locate and extract necessary features needed in the later steps.

*1) Edge detection*

Edge detection was used to detect the edges of the pollen grain in order to locate their position in the image. In this stage, the image input undergoes a series of steps of the algorithm. The resulting image is a binary image that consists the edges and sample image is shown in Figure 5(a).

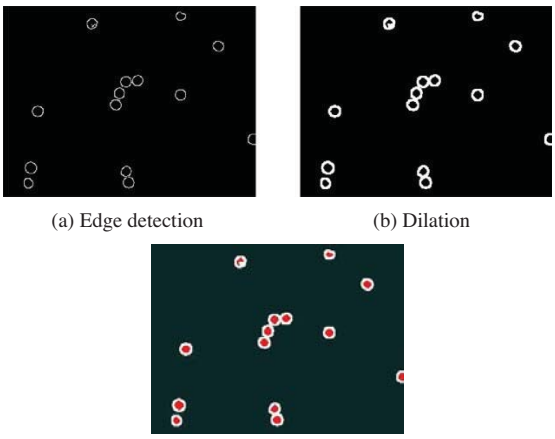
*2) Dilation*

After the application of Canny edge detector, dilation is applied to the image. In this step, the detected edges are

thickened to avoid discontinuities for the improve-ment of the results of region filling which will be applied later. After applying dilation on the binary image produced by the edge detection algorithm, contour tracing is used to locate and count the pollen grains. Figure 5(b) shows a sample image as a result of the dilation performed on the resulting image of the previous process. The coordinates of the pollen grains are stored in a list to be passed as input to Region Filling to fill the regions of the detected pollen grains.

*3) Region filling*

In this step, the regions of the pollen grains' edges are filled as shown in Figure 5(c). Region filling ensures that the unnecessary pixels in the image will be removed since only the parts of the image that forms a region (i.e., the pollen grains) are going to be filled.



(a) Edge detection (b) Dilation

(c) Region filling

Figure 5. Output images result from series of image techniques applied.

## C. Classification of Pollen Grains

To estimate the viability percentage of a given image of pollen samples, there is a need to classify a viable and nonviable pollens. Since staining method visually separates the viable and nonviable pollen grains, this feature was used to identify which pollen grains are viable and which are not. Moreover, considering there might be color variations due to the type of staining applied to the pollen, we convert the image to grayscale to overcome this case. Hence, we'll use the average gray values of each pollen grains to classify them accordingly.

*K*-means clustering algorithm requires as input a dataset and the number of *k* clusters. The dataset will be a collection of the color intensity value of each of the pollen grains. The value of *k* will be 2 since the pollen grains are identified as viable or inviable only. The result will be a clustered version of the data points in the dataset which will be used in the computation of the viability percentage.

# IV. Results and Discussion

In this chapter, the empirical results of the benchmarking of the proposed approach against manual counting is discussed. To evaluate the performance of the image analysis technique, it was tested on a dataset of digital images of microscopic pollen.

## A. Performance of Proposed Pollen Viability Identification

The first experimental setup was carried out to evaluate the performance of the proposed approach for pollen viability identification. Two sets of datasets (with lumped pollen and without lumped pollen) were used. As shown in Table I, an average accuracy of 98.3% was achieved by the system when the dataset without lumped pollen images was used. However, in the case of lumped pollen images, a lower accuracy was achieved averaging only by 85.06%. The drop in accuracy when the images with lumped pollen was used can be attributed to failure of the edge detection method to detect pollens that are overlapping with each other which then causes lesser pollen being identified and classified.

TABLE I. Accuracy of Pollen Grain Classification on Different Sets of Pollen Images

| | **Accuracy (%)** | |
|---|---|---|
| | ***Without Lumped Pollen*** | ***With Lumped Pollen*** |
| Mean | 98.3 | 85.06 |
| Median | 100 | 86.96 |
| Max | 100 | 96.88 |
| Min | 85.37 | 45.83 |
| SD | 4.18 | 10.63 |
| SE | 0.93 | 2.38 |

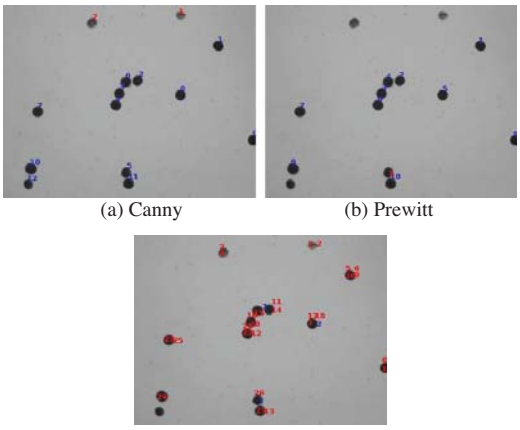
## B. Effect of Varying Edge Detection Algorithm

The next experimental setup aimed to determine the effect of varying edge detection algorithms on the performance of viability classification of the proposed approach. Three commonly known algorithms were used namely Canny, Prewitt, and Sobel. Sample images using the different edge detection algorithms are shown in Figure 6. As observed in Table II, on the set of images without lumped pollen grains, Canny edge detector yielded best results with 98.38% accuracy. Next is Prewitt edge detector with 67.47% accuracy and Sobel edge detector had the least accuracy with only 40.74%. Sobel had the most dissimilarities with the results obtained manually. Canny and Prewitt were able to detect the pollen grains well. However, Canny still counted the pollen grains better compared to Prewitt.

TABLE II. Accuracy of Pollen Grain Classification with Varying Edge Detection Algorithms on the Set of Pollen Images without Lumped Pollen Grains

| | **Accuracy (%)** | | |
|---|---|---|---|
| | ***Canny*** | ***Prewitt*** | ***Sobel*** |
| Mean | 98.38 | 67.47 | 40.74 |
| Median | 100.00 | 75.60 | 32.10 |
| Max | 100.00 | 100.00 | 92.86 |
| Min | 85.37 | 21.74 | 9.52 |
| SD | 4.18 | 25.80 | 26.02 |
| SE | 0.93 | 5.77 | 5.82 |

The main reason for the superiority of the the Canny operator can be attributed to additional steps performed to make sure that noise is removed and only valid edges are detected thus making it able to detect the pollen grains well. The Prewitt operator is fairly good in detecting clean edges, however, light edges get easily suppressed from the thresholding making it count less accurate compared to Canny. On the other hand, the Sobel operator detects overly disconnected edges that were impossible to be connected using dilation, thus Sobel's counts were the least accurate among the three edge detection operators.



(a) Canny (b) Prewitt

(c) Sobel

Figure 6. Sample image applied with varying edge operators.

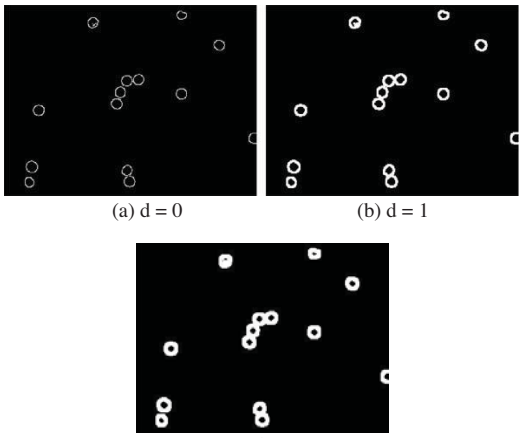
## C. *Effect of Varying Dilation Values*

In this experiment, different dilation values were tested to determine which dilation value is ideal. This experiment was conducted to determine the ideal value to connect the disconnected edges from the edge detection step without eliminating important details of the pollen grains. The results of the accuracy on the classification of pollen grains with varying dilation values are shown in Table III.

When the dilation is set to 0 (Figure 7(a)), the disconnected edges don't get to form regions which result for it to be not detected and counted as pollen grain in the later steps of the protocol. The same thing happens when dilation is set to 2 (Figure 7(c)). The dilation becomes too thick that it covers the whole region of the pollen grain. The three operators detected the pollen grains well when the dilation is set to 1 (Figure 7(b)) since it connected the edges just enough without losing too many details. Canny, with dilation=1 counted the pollen grains better on the set of images without lumped pollen grains with 98.38% accuracy.

TABLE III. ACCURACY OF POLLEN GRAIN CLASSIFICATION WITH VARYING DILATION VALUES ON THE SET OF POLLEN IMAGES WITHOUT LUMPED POLLEN GRAINS

| | Accuracy (%) | | |
|---|---|---|---|
| | ***d=0*** | ***d=1*** | ***d=2*** |
| Mean | 90.56 | 98.38 | 95.41 |
| Median | 92.27 | 100.00 | 100.00 |
| MAX | 100.00 | 100.00 | 100.00 |
| MIN | 73.17 | 85.37 | 72.22 |
| Var | 90.79 | 17.46 | 55.55 |
| SD | 9.53 | 4.18 | 7.45 |
| SE | 2.13 | 0.93 | 1.67 |



(a) d = 0 (b) d = 1

(c) d = 2

Figure 7. Sample images of varying dilation values.

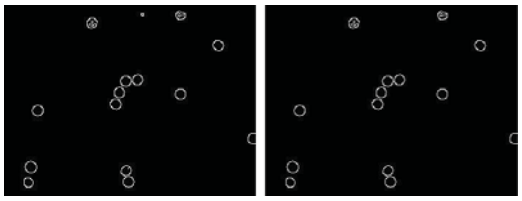
## D. *Effect of Varying Threshold Values*

The third and final experimental setup is varying the threshold values of Canny edge detection algorithm. This experiment determines which threshold value most affect the accuracy of the classification of the pollen grains on the different sets of pollen images. The test was conducted with the configuration of dilation=1 and Canny for the edge detection.

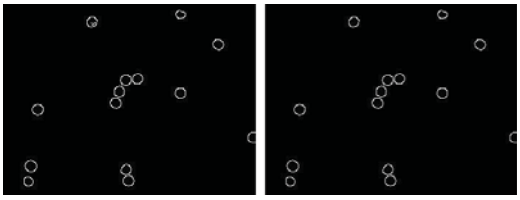
TABLE IV. ACCURACY OF POLLEN GRAIN CLASSIFICATION WITH VARYING THRESHOLD VALUES ON THE SET OF POLLEN IMAGES WITHOUT LUMPED POLLEN GRAINS

| | Accuracy (%) | | | |
|---|---|---|---|---|
| | ***t=30*** | ***t=50*** | ***t=70*** | ***t=85*** |
| Mean | 95.75 | 97.22 | 98.74 | 97.83 |
| Median | 97.01 | 100.00 | 100.00 | 100.00 |
| MAX | 100.00 | 100.00 | 100.00 | 100.00 |
| MIN | 85.71 | 85.71 | 87.50 | 87.50 |
| Var | 23.95 | 22.24 | 10.13 | 12.75 |
| SD | 4.89 | 4.72 | 3.18 | 3.57 |
| SE | 1.09 | 1.05 | 0.71 | 0.80 |

On the set of images without lumped pollen grains, results shown in Table IV of all the tested threshold values produced fairly similar results. When the threshold is set to 30-50, the pollen grains were classified correctly. However, there are still noise detected in the background (Figure 8(a) and 8(b)). Though the noise will be removed in the later steps of the protocol, there might still be a case that a noisier image will be mistaken as a pollen grain. There was least noise detected when the threshold value is set from 70-85 (Figure 8(c) and 8(d)) with 98.74% and 97.83% accuracy respectively. When the threshold value is from 70-85, more details of the pollen grain were preserved making the system able to detect and count pollen grains more accurately.



(a) threshold=30 (b) threshold=50



(c) threshold=70 (d) threshold=85

Figure 8. Sample image applied with varying threshold values.

## V. CONCLUSION

An image processing technique on pollen viability identification on selected rice genotypes was proposed in this study. A series of image processing methods were applied to the dataset of microscopic pollen images of rice. Edge detection technique, dilation, and region filling was applied to detect the pollen in the input image. Different edge detection algorithms were used namely Canny, Prewitt, and Sobel. It was then classified using unsupervised learning method using *k*-Means clustering to classify the viable and non-viable pollen. The performance of the proposed technique was evaluated by measuring the accuracy of the classification of pollens.

Empirical results have shown that the performance of the proposed approach for pollen viability classification is comparable to manual counting. These results suggest that the presented image analysis protocol can provide an alternative to more labor-intensive methods for estimating pollen viability. The system provided a fast, inexpensive and reliable solution to counting and computing pollen viability percentage from digital images. The system has also computed the viability percentage of a dataset of pollen images in an average of 8 seconds per image, considerably faster than the manual method which takes 10-60 minutes.

The current proposed approach had difficulty on classifying pollen that are overlapping with each other. Further study is suggested in improving the system in identifying the pollen grain on images with lumped pollen. Adding machine learning can help the system be more dynamic in the identification of the pollen grains as well as the parameter values used in other algorithms.

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